Advanced Calculus

Continuous Functions

Continuity • Uniform Continuity

連續性 • 均勻連續

Yu-Xuan Chen

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§ 3.3

Continuity

連續性

Definition: Continuity

Definition 3.19

Let $E \subseteq \mathbb{R}$ be nonempty and $f : E \to \mathbb{R}$.

i) f is **continuous at** $a \in E$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - a| < \delta \text{ and } x \in E \implies |f(x) - f(a)| < \varepsilon.$$

ii) f is **continuous on** E if it is continuous at every $a \in E$.

Compare with Definition 3.1 (limit)

- L is replaced by $f(a)$, so f **must be defined at** a.
- No "$0 < |x - a|$" : the point $x = a$ is allowed (it gives $0 < \varepsilon$ trivially).
- Only $x \in E$ matters, so E need not be open: e.g. $\sqrt{x}$ is continuous at $0 \in [0, \infty)$.

連續的定義只看定義域 E 裡的點，所以端點也可以談連續（只看單邊）。

Equivalent Forms of Continuity

Remark 3.20 (open interval)

If I is an open interval containing a and $f : I \to \mathbb{R}$, then

$$f \text{ is continuous at } a \iff \lim_{x \to a} f(x) = f(a).$$

Theorem 3.21 (Sequential Characterization of Continuity)

For $a \in E$, the following are equivalent:

i) f is continuous at a;

ii) for every sequence $x_n \in E$ with $x_n \to a$, we have $f(x_n) \to f(a)$.

Checklist: f continuous at an interior point a means three things:

(1) $f(a)$ defined (2) $\lim_{x \to a} f(x)$ exists (3) the two are equal.

和 Theorem 3.6 不同：這裡 x_n 可以等於 a。

Building Continuous Functions

Theorem 3.22 (Algebra)

If f, g are continuous at a (or on E), then so are

$$f+g, \qquad fg, \qquad \alpha f, \qquad \frac{f}{g}\ (g(a)\neq 0).$$

Also $|f|$, f^+, f^-, $f\vee g=\max\{f,g\}$, $f\wedge g=\min\{f,g\}$. Every polynomial is continuous on $\mathbb{R}$.

Theorem 3.24 (Composition)

i) If $\lim_{x\to a} f(x)=L$ and g is continuous at L, then $\lim_{x\to a} g\big(f(x)\big)=g\Big(\lim_{x\to a} f(x)\Big)$.

ii) If f is continuous at a and g is continuous at $f(a)$, then $g\circ f$ is continuous at a.

3.24 (i) 是「極限與函數值可以交換」的第一個例子：g 連續時，$\lim$ 可以搬進 g 裡面。

Extreme Value Theorem

Definition 3.25

f is **bounded** on E if there is M with $|f(x)| \leq M$ for all $x \in E$.

Theorem 3.26 (Extreme Value Theorem, EVT)

If I is a **closed, bounded** interval and $f : I \to \mathbb{R}$ is continuous, then f is bounded on I, and there are $x_m, x_M \in I$ with

$$f(x_M) = \sup_{x \in I} f(x), \qquad f(x_m) = \inf_{x \in I} f(x).$$

Remark 3.27: neither hypothesis can be dropped

- Bounded, not closed: $f(x) = \frac{1}{x}$ on $(0, 1)$ is unbounded.
- Closed, not bounded: $f(x) = x$ on $[0, \infty)$ is unbounded.

最大值、最小值「取得到」：$\sup$ 和 $\inf$ 都是某一點的函數值。

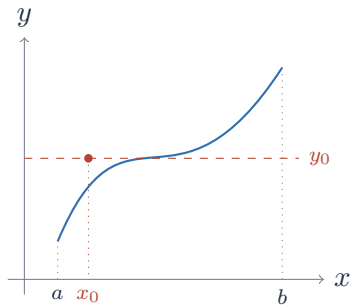
Intermediate Value Theorem

Theorem 3.29 (IVT)

If $f : [a, b] \to \mathbb{R}$ is continuous and y_0 lies between $f(a)$ and $f(b)$, then there is $x_0 \in (a, b)$ with $f(x_0) = y_0$.

Lemma 3.28 (sign persistence)

If f is continuous at x_0 and $f(x_0) > 0$, then $f > \varepsilon > 0$ on some $[x_0, x_1]$.



Proof idea (technique worth remembering): let $E = \{x \in [a, b] : f(x) < y_0\}$ and $x_0 = \sup E$; continuity + Lemma 3.28 force $f(x_0) = y_0$.

- Typical use: show an equation $h(x) = 0$ has a root — apply IVT to h with $y_0 = 0$.

IVT 用來證明「解存在」：找兩點使 h 一正一負即可。

Examples of Discontinuity

不連續可以有多「壞」？

	$f(x)$	behavior at the bad point(s)
Ex 3.30	$\frac{\lvert x\rvert}{x}$ $(x \neq 0)$, $f(0) = 1$	jump: $f(0+) = 1 \neq -1 = f(0-)$
Ex 3.31	$\sin \frac{1}{x}$ $(x \neq 0)$, $f(0) = 1$	neither $f(0+)$ nor $f(0-)$ exists
Ex 3.32	Dirichlet: 1 on $\mathbb{Q}$, 0 off $\mathbb{Q}$	discontinuous **everywhere**
Ex 3.33	$\frac{1}{q}$ at $x = \frac{p}{q}$, 0 at irrationals	continuous only at irrationals in $(0, 1)$

Remark 3.34

$g \circ f$ can be nowhere continuous even if f is discontinuous only on $\mathbb{Q}$ and g is discontinuous at only one point.

Ex 3.33（Thomae 函數）：在有理點用「無理數列」逼近就能看出不連續。

Proof Techniques for Continuity

Showing f is continuous

- Break f into sums / products / quotients / compositions of known continuous functions (Theorems 3.22, 3.24).
- At a gluing point of a piecewise function: check $f(a+) = f(a-) = f(a)$.
- Directly by ε–δ when the problem asks for it.

Showing f is NOT continuous at a

- Find $x_n \to a$ with $f(x_n) \not\to f(a)$ (Theorem 3.21).
- Density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$: every interval contains both kinds of points (used in Ex 3.32, 3.33).

Using continuity

f continuous and $x_n \to a \Rightarrow f(x_n) \to f(a)$: pass limits through f.

§ 3.3 Common Mistakes 常見錯誤

Mistake 1: limit exists $\Rightarrow$ continuous

Also need $f(a)$ defined **and** equal to the limit. E.g. $f(x) = x$ $(x \neq 0)$, $f(0) = 1$: $\lim_{x \to 0} f(x) = 0 \neq f(0)$.

Mistake 2: forgetting the hypotheses of EVT

"f is continuous on $(0, 1)$, so it attains its maximum." EVT needs a **closed and bounded** interval.

Mistake 3: using IVT without continuity

$f(x) = \frac{|x|}{x}$ $(f(0) = 1)$ takes -1 and 1 on $[-1, 1]$ but never 0.

Mistake 4: $\frac{1}{x}$ is "discontinuous at 0"

$0 \notin \mathrm{Dom}(f)$, so continuity at 0 is not even defined; $\frac{1}{x}$ is continuous on its domain.

§3.4

Uniform Continuity

均勻連續

Definition: Uniform Continuity

Definition 3.35

$f : E \to \mathbb{R}$ is **uniformly continuous** on E if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - a| < \delta \text{ and } x, a \in E \implies |f(x) - f(a)| < \varepsilon.$$

The only difference: order of quantifiers

$$\text{continuous on } E: \quad \forall a \in E \ \forall \varepsilon > 0 \ \exists \delta = \delta(\varepsilon, a) > 0 \ \forall x \in E \dots$$
$$\text{uniformly continuous on } E: \quad \forall \varepsilon > 0 \ \exists \delta = \delta(\varepsilon) > 0 \ \forall a, x \in E \dots$$

- Uniformly continuous $\Rightarrow$ continuous; the converse can fail.
- Uniform continuity is a property **on a set**, not at a point.

均勻連續：同一個 δ 要對整個 E 的所有點都有效。證明時要在提到 a **之前**就決定 δ。

Proving Uniform Continuity Directly

Example 3.36: $f(x) = x^2$ on $(0, 1)$

1. Factor: $|x^2 - a^2| = |x - a|\,|x + a|$.
2. Bound the other factor **uniformly in** a: for $x, a \in (0, 1)$, $|x + a| \le 2$.
3. So $|f(x) - f(a)| \le 2|x - a|$; take $\delta = \varepsilon/2$.

General pattern: Lipschitz-type estimate

If $|f(x) - f(a)| \le C\,|x - a|$ for all $x, a \in E$ with a constant C independent of x, a, then f is uniformly continuous on E (take $\delta = \varepsilon/C$).

和 §3.1 找 δ 的手法一樣，只是「另一個因子」的上界不能依賴 a；有界的定義域讓這件事做得到。

Disproving Uniform Continuity

Negation

$\exists \varepsilon_0 > 0 \ \forall \delta > 0 \ \exists x, a \in E : \ |x - a| < \delta$ but $|f(x) - f(a)| \geq \varepsilon_0$.

Sequence version: find $x_n, y_n \in E$ with

$$|x_n - y_n| \to 0 \qquad \text{but} \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

Example 3.37: x^2 on $\mathbb{R}$

Take $a = n$, $x = n + \frac{\delta}{2}$:

$$|x^2 - a^2| = n\delta + \frac{\delta^2}{4} > n\delta.$$

Large n makes this > 1.

$\frac{1}{x}$ on $(0, 1)$

$x_n = \frac{1}{n}$, $y_n = \frac{1}{n+1}$:

$$|x_n - y_n| \to 0, \quad |f(x_n) - f(y_n)| = 1.$$

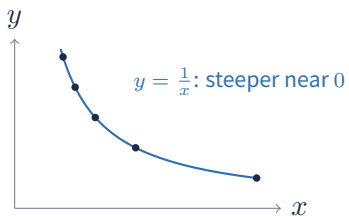
兩種「壞」：定義域無界讓斜率越來越大（x^2），或函數在有限端點附近爆掉（$\frac{1}{x}$）。

Cauchy Sequences and Uniform Continuity

Lemma 3.38

If $f : E \to \mathbb{R}$ is uniformly continuous and $\{x_n\} \subseteq E$ is Cauchy, then $\{f(x_n)\}$ is Cauchy.

- **Test for non-uniform continuity:** find a Cauchy $\{x_n\}$ with $\{f(x_n)\}$ not Cauchy.
- Example: $f(x) = \frac{1}{x}$ on $(0, 1)$, $x_n = \frac{1}{n}$ is Cauchy but $f(x_n) = n$ is not.
- Continuity alone does **not** preserve Cauchy sequences.



Closed, Bounded Intervals

Theorem 3.39

If I is a **closed, bounded** interval and $f : I \to \mathbb{R}$ is continuous, then f is uniformly continuous on I.

Proof structure (same pattern as EVT):

not u.c. $\Rightarrow$ $x_n, y_n \in I$, $|x_n - y_n| < \frac{1}{n}$, $|f(x_n) - f(y_n)| \geq \varepsilon_0$

$\overset{\text{BW}}{\Longrightarrow}$ subsequences converge to the same point $\Rightarrow$ contradiction

Theorem 3.40 (bounded open intervals)

$f : (a, b) \to \mathbb{R}$ is uniformly continuous on (a, b) $\iff$ f extends to a continuous $g : [a, b] \to \mathbb{R}$.

Practical test: for continuous f on bounded (a, b): f u.c. $\iff$ $f(a+)$ and $f(b-)$ exist (finite).

3.40 對無界區間**不成立**；有界開區間只要檢查兩端點的單邊極限。

Using Theorem 3.40

Example 3.41: $f(x) = \frac{x-1}{\log x}$ **on** $(0,1)$

$$\lim_{x\to 0^+} f(x) = 0, \qquad \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \frac{1}{1/x} = 1 \ \text{(l'Hôpital)}.$$

Both one-sided limits exist $\Rightarrow$ f extends continuously to $[0,1]$ $\Rightarrow$ u.c. on $(0,1)$.

Quick classification on $(0,1)$

$f(x)$	$\lim_{x\to 0^+} f(x)$	u.c. on $(0,1)$?
x^2	0	yes
$x \sin \frac{1}{x}$	0	yes
$\frac{1}{x}$	∞	no
$\sin \frac{1}{x}$	does not exist	no

$\sin \frac{1}{x}$ 有界但不均勻連續：有界不夠，端點的極限一定要存在。

§ 3.4 Common Mistakes 常見錯誤

Mistake 1: δ depends on the point

"Take $\delta = \varepsilon/(2|a|)$." For uniform continuity δ may depend only on ε (and f, E).

Mistake 2: continuous on a bounded set $\Rightarrow$ u.c.

$\frac{1}{x}$ on $(0,1)$ The interval must be **closed** and bounded (Theorem 3.39).

Mistake 3: bounded $\Rightarrow$ u.c.

$\sin\frac{1}{x}$ is bounded on $(0,1)$ but not uniformly continuous.

Mistake 4: applying Theorem 3.40 on unbounded intervals

x^2 on $(0,\infty)$: no endpoint problem at 0, yet it is not u.c. (Example 3.37).

One-Page Summary

Continuity	$\forall \varepsilon\ \exists \delta(\varepsilon, a):\ \lvert x-a\rvert < \delta,\ x \in E \Rightarrow \lvert f(x)-f(a)\rvert < \varepsilon$
Thm 3.21	continuous at $a \iff f(x_n) \to f(a)$ for all $x_n \to a$ in E
Thm 3.22, 3.24	sums, products, quotients, compositions stay continuous
EVT (3.26)	continuous on closed bounded $I \Rightarrow$ max and min attained
IVT (3.29)	continuous on $[a,b] \Rightarrow$ takes every value between $f(a)$, $f(b)$
Uniform cont.	$\forall \varepsilon\ \exists \delta(\varepsilon):\ \lvert x-a\rvert < \delta,\ x,a \in E \Rightarrow \lvert f(x)-f(a)\rvert < \varepsilon$
Lemma 3.38	u.c. maps Cauchy sequences to Cauchy sequences
Thm 3.39, 3.40	$[a,b]$: cont. $\Rightarrow$ u.c.; (a,b): u.c. $\iff$ $f(a+), f(b-)$ exist

Watch out: closed & bounded for EVT / 3.39 $\quad$ δ independent of a $\quad$ bounded $\not\Rightarrow$ u.c.